B. Specific Aims: Discrimination, Grouping, and Segmentation of Natural Texture

Visual systems are matched (via evolution and learning over the lifespan) to the tasks the organism performs to survive and reproduce. Thus, it is of fundamental importance to analyze visual systems with respect to natural tasks and to the statistical properties of natural stimuli relevant to performing those tasks. In our lab we call this "natural systems analysis." It is composed of several steps: (1) identify natural tasks, (2) measure the natural scene statistics relevant for those tasks, (3) determine how to optimally use those statistics to perform the tasks, given appropriate biological constraints, and (4) use the first three steps to formulate principled hypotheses which are tested and refined in behavioral or physiological experiments. We believe this approach is critically important to progress in vision science.

Here, we propose to focus on the fundamental task of perceptual grouping and segmentation of natural and naturalistic textures. The human visual system has a remarkable ability to rapidly group and segment natural scenes into regions that most likely contain the same physically meaningful content. It can do this well with arbitrary new scenes, even though the resolution of the visual system declines rapidly away from the line of sight. At the core of this ability is the ability to identify whether small patches of natural texture are samples from the same or different types of texture, even for textures that have never been seen in the past. This same-different discrimination ability is an essential and powerful factor supporting the visual system's ability to correctly interpret natural images. Although there is a long history of vision research on perceptual grouping and segmentation, there have been no direct measurements of how well humans can identify whether pairs of random natural texture patches are the same or different. Nor do we know if there are theories that can predict human performance in such tasks. Our first two aims are to measure and model texture-content discrimination and texture-boundary detection for small natural texture patches the size of typical cortical receptive fields. Our third aim is to build upon the first two aims by measuring and modeling human ability to group and segment images composed of natural texture patches.

Aim 1: Discrimination of Separated Natural Texture Regions. Same-different texture discrimination will be measured as a function of retinal eccentricity, for pairs of spatially separated 1° x 1° gray-scale texture patches randomly sampled from existing natural texture databases. We will develop and test two classes of image-computable model. One class is Bayesian observers based directly on measurements of natural image statistics and known properties of the visual system (e.g., the optics of the eye and the retinal-ganglion-cell sampling mosaic). The models will include multiple specific image properties. We have developed an efficient method of characterizing the joint statistical distributions of these image properties across a wide range of natural textures, which allows us to make principled predictions for the same-different task. In preliminary studies, we have found that these Bayesian observers can accurately perform the same-different task with a single fixed decision bound that is independent of the specific textures. The other class of models will be self-supervised convolutional neural networks (SSCNNs) trained on the same-different task. We have developed a new self-supervised training procedure that is more biologically plausible and flexible than training with labelled datasets. One potential value of the SSCNNs is that they may discover computations that capture human ability to make accurate same-different judgments across a wider range of conditions. Different versions of the Bayesian and SSCNN observers (as well as other earlier models) will be compared with human performance.

Aim 2: Discrimination of Adjacent Natural Texture Regions. Texture discrimination of adjoining regions will be measured as a function of retinal eccentricity, in a similar paradigm, for two different cases. In one case (maximum camouflage), the texture content will be the same across the boundary: a 1° x 2° patch of the texture or two abutting 1° x 1° random samples of the same texture. Here, most of the discrimination information between the two regions is in the boundary cues. In the second case, the texture patches on the different trials will be from different textures. Here there is discrimination information in both the boundary and the content differences. Again, human performance will be compared with Bayesian and SSCNN observer models.

Aim 3: Grouping Natural Texture Regions. Grouping and segmentation of texture regions will be measured in a "rows or columns" (RC) task, where observers judge the dominant organization (rows or columns) in random arrays of patches (from two different randomly selected textures). The proposed image-computable models will combine the same-different discrimination models above with four principles of perceptual grouping: similarity, proximity, good continuation, and "mutual similarity" (a new proposed grouping principle). In preliminary studies we implemented a version of this class of models and found that all four principles contribute strongly to model performance in the RC task.